Correlated Exploration for Cooperative MARL

via a Gaussian Copula

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Project 9 — Reinforcement Learning

The problem: coordinated exploration

In cooperative MARL the team is often rewarded **only** when agents act *together* (e.g. both reach a goal in the same step).

- Standard ε-greedy explores each agent **independently**
- Probability all n agents explore the matching joint action $\propto \varepsilon^n$
- $\rightarrow$ decays **exponentially** in the number of agents

Hypothesis: coupling agents' exploration — so similar agents tend to act alike — should accelerate learning on coordination-bottlenecked tasks.

Method: Gaussian copula

Keep the ε-greedy template, but **correlate which action** the exploring agents take (who explores stays an independent Bernoulli):

1. Build a correlation matrix R (PSD, unit diagonal)
2. Cholesky $R = LL^\top$
3. Sample $z \sim \mathcal{N}(0, I)$, set $y = Lz \sim \mathcal{N}(0, R)$
4. $u_i = \Phi(y_i)$ → uniform marginals, **correlated** across agents

An exploring agent takes action $\lfloor u_i \cdot |A_i| \rfloor$. Correlated agents draw similar u_i → **same joint action** (focus-fire, both on switches). Independent ε-greedy is the special case $R = I$.

Where does the correlation come from?

R = **cosine similarity** of per-agent feature vectors $\rightarrow$ a Gram matrix, automatically PSD with unit diagonal (+ small jitter).

Three similarity sources compared:

- `obs` — the agent's observation
- `q_values` — its current Q-values
- `hidden` — its Q-network backbone activations

Role-agnostic: nothing tells the method which agents share a role — similar agents simply get correlated automatically.

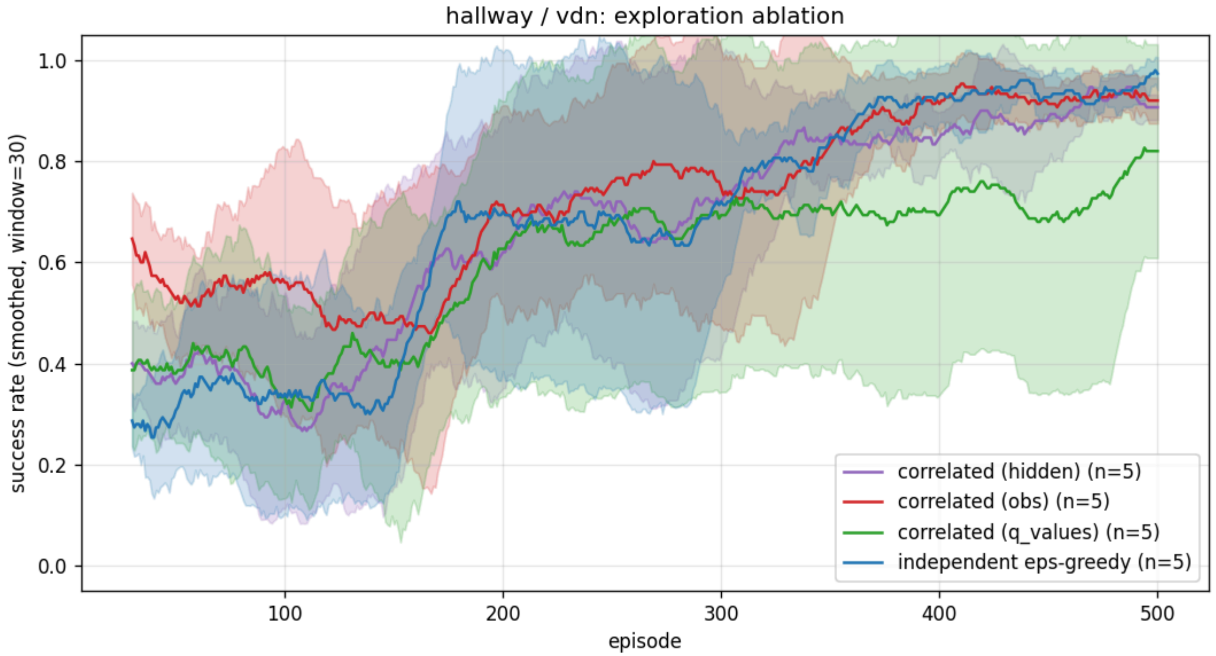
Experimental setup

Environments

- **Hallway** (MAVEN) — 2 agents, separate corridors, reward only if both reach 0
- **Level-Based Foraging** — 2 agents must load food *together* (sparse)
- **SMAX** `2s3z` (StarCraft-like, JAX) — 5 agents, 2 stalkers + 3 zealots

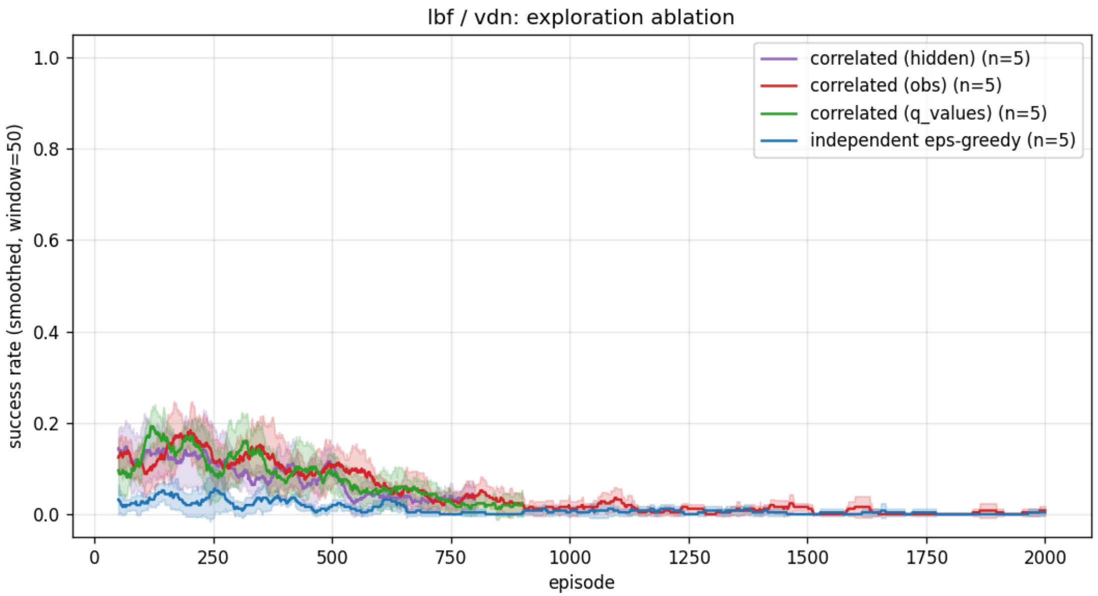
Algorithms: VDN (sum) and QMIX (monotonic mixer), parameter-shared Q-network, DQN-style training, **5 seeds** each.

Result 1 — Hallway: faster early learning



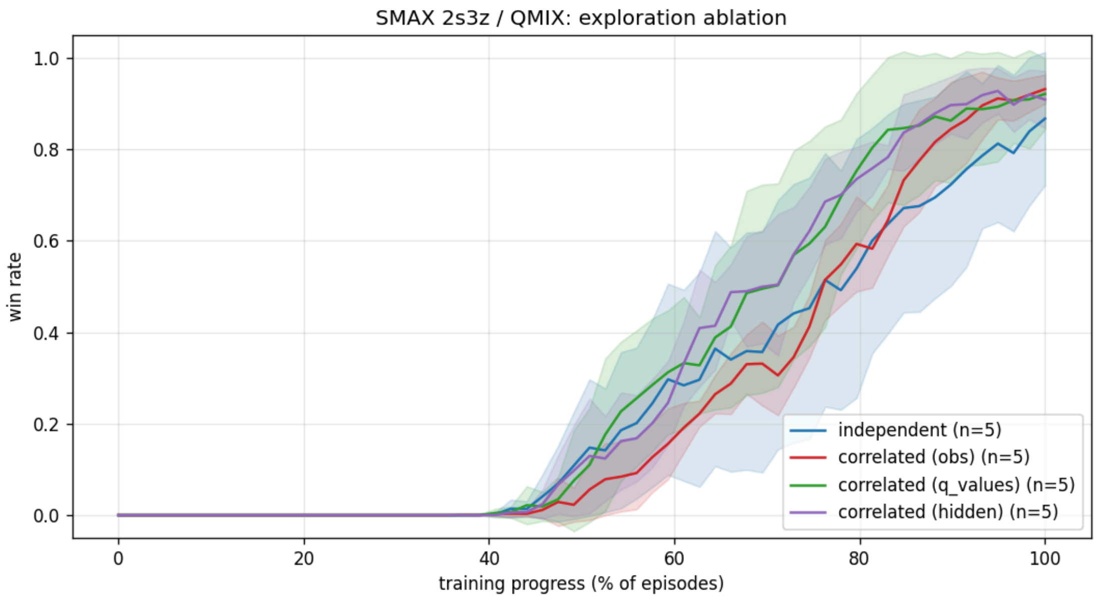
Correlated-`obs` early success (ep 50–100): **+64%** (VDN), **+94%** (QMIX) over independent. All converge eventually → the benefit is a **speed-up**.

Result 2 — LBF: the mechanism works even where learning fails



Catastrophic forgetting collapses all methods, **but** correlated reaches a **2.6–2.9× higher peak** — it finds the coordinated successes independent exploration misses.

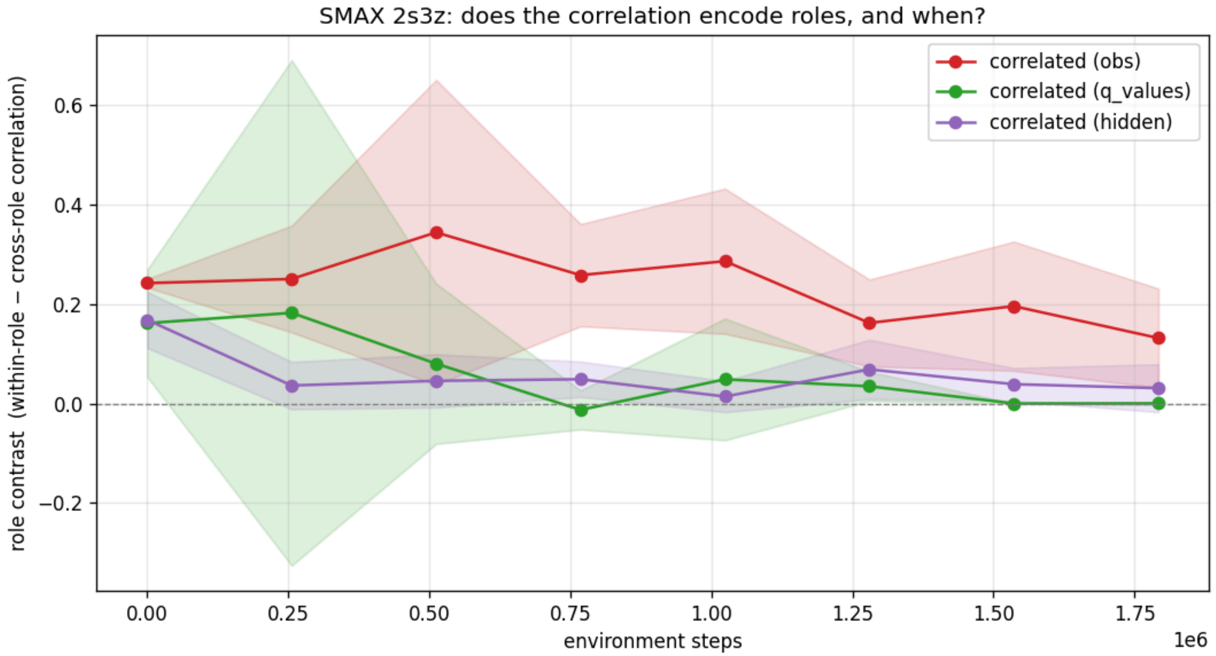
Result 3 — SMAX 2s3z: scales to GPU



Under **QMIX**, correlated lifts the **final win rate 0.81 → 0.91** (and reaches 50% sooner); under the easier VDN all reach ~0.95, correlated ~6% faster.

(Convergence required Double DQN + soft targets + a LayerNorm on the SMAX state.)

Result 4 — *when* does correlation encode roles?



Role contrast = within-role − cross-role correlation. **`obs`** stays positive through the exploration-heavy phase (always encodes unit type); **`q_values`** is noisy, **`hidden`** weak → why `obs` is the reliable source.

Conclusions

- Correlating agents' exploratory **actions** via a Gaussian copula is a **simple, drop-in** change to exploration
- **Accelerates** learning where coordination is the bottleneck (Hallway +64–94%)
- **Improves exploration quality** even where value learning fails (LBF 2.6–2.9×)
- **Scales to GPU (SMAX):** lifts QMIX final win rate **0.81** → **0.91**
- `obs` is the most reliable similarity source; couples situationally-similar agents (stalkers most strongly) with no role info supplied

Thank you — questions?